



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to SRI RAMA ANJANEYA FUELS

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

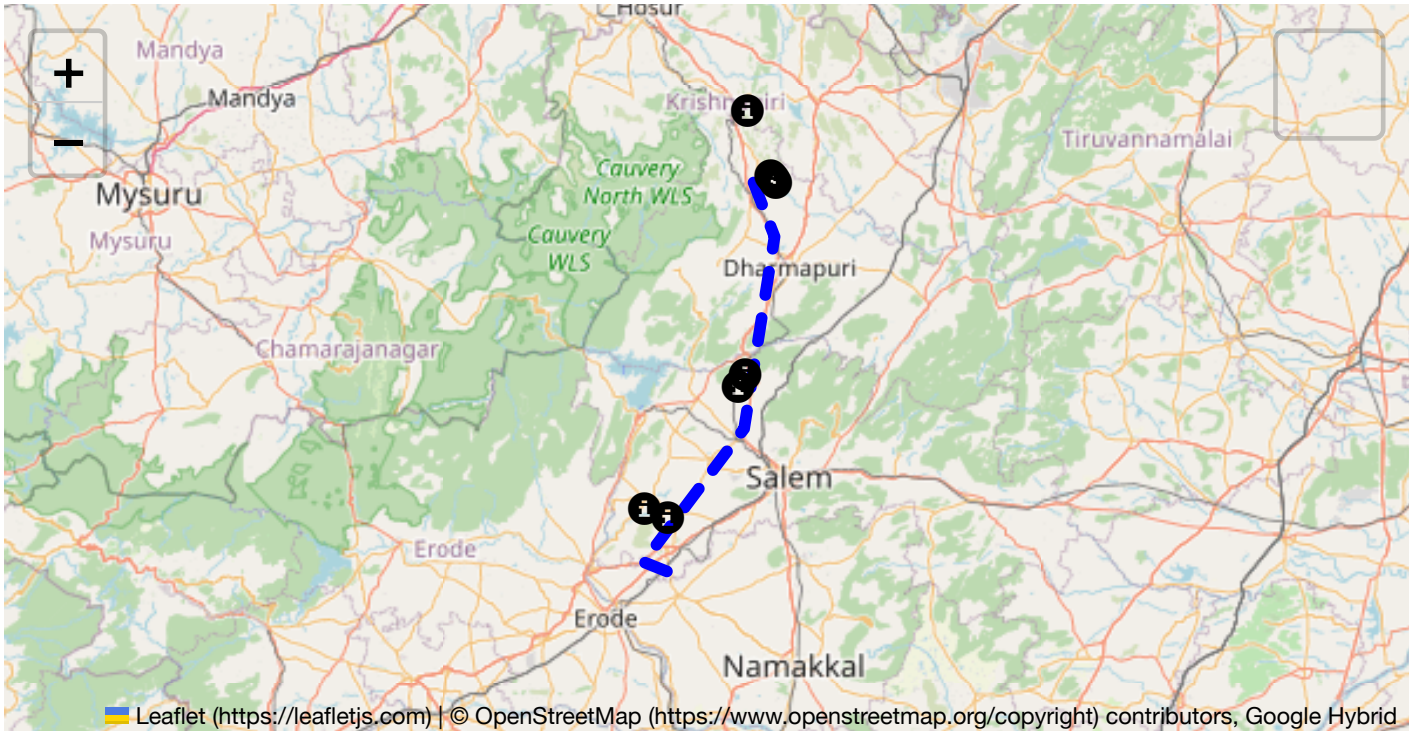
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

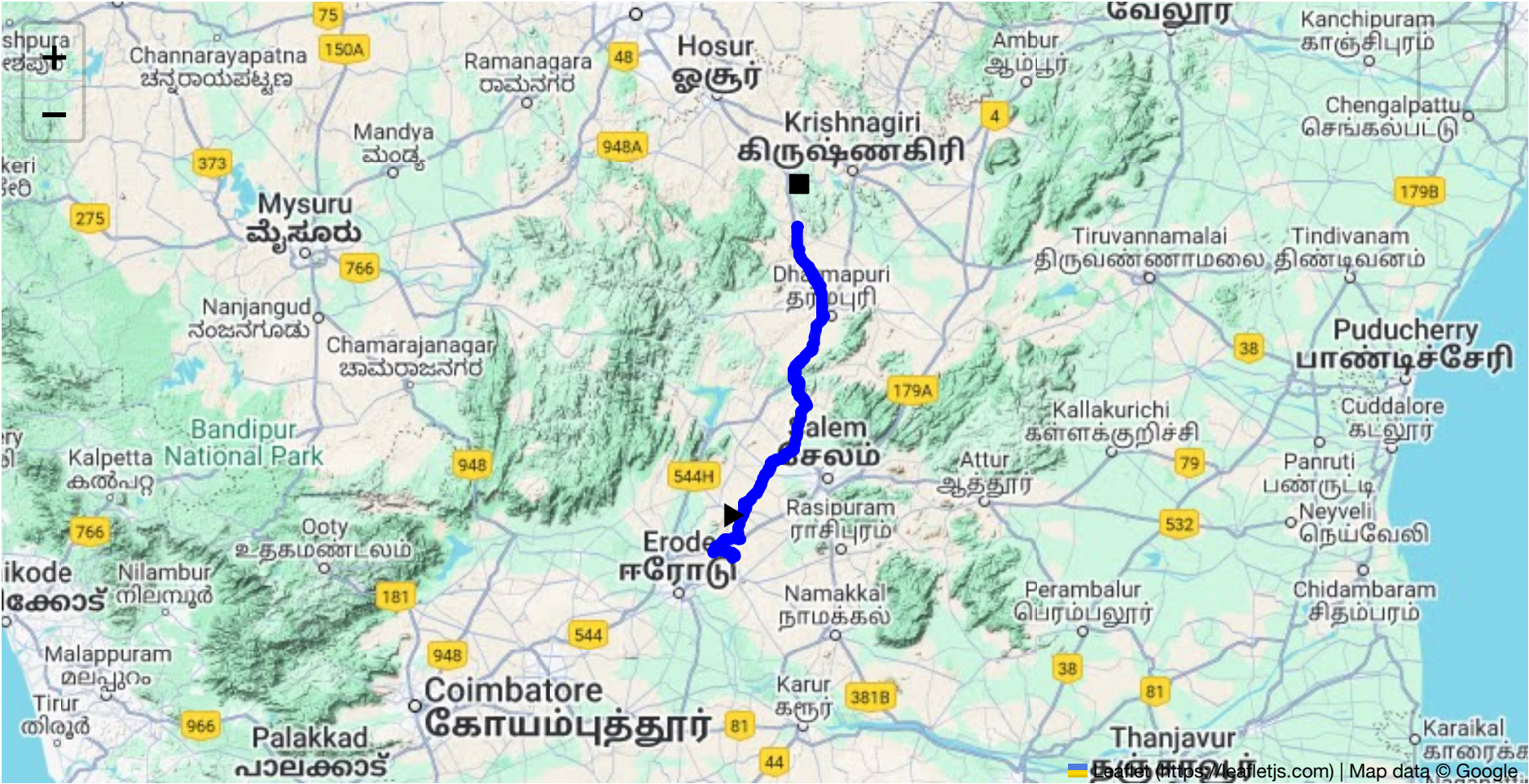
Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 138.04 km
Estimated Duration: 2.8 hours
Adjusted Duration (Heavy Vehicle): 3.5 hours

Start: (11.4381, 77.8734)
End: (12.369807, 78.060169)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map:

The route from Sangagiri to Golasanahalli is approximately 138.04 kilometers and involves traveling through major towns like Salem, Omalur, and Dharmapuri. The journey begins in Sangagiri, goes north through Bhavani Main Road, continues through Salem, Omalur, and Dharmapuri, and ends in Golasanahalli. The landscape includes both urban and semi-urban areas, with significant stretches of rural roads.

2. Typical Weather Conditions and Potential Weather-Related Hazards:

The region experiences a tropical climate with hot summers, moderate rainfall during the monsoon, and mild winters. Weather-related hazards include:

- Heavy rains and flooding during the southwest monsoon (June to September), which may cause road closures and reduced visibility.
- Heatwaves during peak summer months (April to June), which could lead to vehicle overheating.

3. Analysis of Traffic Patterns:

- *Peak Hours and Congestion-Prone Areas:*
 - Salem, particularly the city center and entry/exit points, faces high congestion during morning (8:30-10:30 AM) and evening (5:30-7:30 PM) peak hours.
 - Omalur to Dharmapuri stretch can also experience heavy truck traffic, particularly near industrial zones.

4. Assessment of Road Quality and Infrastructure:

- Roads in and around Salem are well-maintained, but some rural stretches may have narrower lanes and occasional potholes.

- Infrastructure is generally sufficient for heavy vehicles; however, maintenance checks are suggested post-monsoon for pothole repairs.

5. Suggestions for Alternative Routes:

- During emergencies, a diversion through State Highway 20 (SH 20), which runs parallel to sections of the main route, may be considered.
- Utilize local roads connecting back to National Highway 44 (NH 44) for northbound alternatives.

6. Summary of Local Regulations Affecting Hazardous Material Transport:

- Adherence to state regulations regarding the transport of hazardous materials is crucial. This includes mandatory permits and adherence to prescribed routes and timing restrictions.
- Vehicles should be equipped with safety equipment, including fire extinguishers and spill kits.

7. Overview of Historical Incidents:

- Salem and sections around Dharmapuri have reported few incidents involving heavy vehicles, mostly due to overloading or faulty vehicle conditions.
- There is no significant history of hazardous material incidents, but caution is advised given the proximity to industrial areas.

8. Environmental Considerations and Sensitive Areas:

- Sections near Dharmapuri and Salem are close to environmentally sensitive zones, including agricultural land. Drivers should avoid spillage and follow spill response protocols.
- Noise and emissions should be minimized, especially in populated areas.

9. Analysis of Communication Coverage:

- Urban areas like Salem have good network coverage, whereas rural stretches may experience intermittent connectivity issues, notably around Pulikkarai.
- Ensure you have a communication plan that includes offline maps and emergency contact protocols.

10. Estimated Emergency Response Times:

- In urban areas like Salem, response times are generally quick (10-20 minutes).
- Rural areas may experience delays of up to 30-45 minutes due to distance and road conditions.

11. An Overall Summary of Risk Assessment:

The route is moderately challenging due to mixed road conditions, weather impact during monsoons, and potential congestion spots. Rigorous adherence to safety protocols for transporting hazardous materials is advised. Drivers should remain informed about weather forecasts, follow regulated routes, and have contingency plans for emergency situations. Regular vehicle maintenance and checks are crucial, especially before the monsoon season, to mitigate the risks of accidents due to road and weather conditions. Overall, with appropriate precautions, the route is manageable for heavy vehicles carrying hazardous materials.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43968, 77.87345	15 KM/Hr
1	Turn	High	11.44029, 77.87544	15 KM/Hr
2	Turn	High	11.44900, 77.87406	15 KM/Hr
3	Turn	High	11.45353, 77.85697	15 KM/Hr
4	Turn	Medium	11.45781, 77.85203	30 KM/Hr
5	Turn	High	11.46307, 77.84941	15 KM/Hr
6	Turn	High	11.45542, 77.81725	15 KM/Hr
7	Turn	High	11.45605, 77.81683	15 KM/Hr
8	Blind Spot	Blind Spot	11.49236, 77.89552	10 KM/Hr
9	Turn	Medium	11.49365, 77.88596	30 KM/Hr
10	Turn	High	11.49465, 77.88488	15 KM/Hr
11	Turn	High	11.55608, 77.89762	15 KM/Hr
12	Turn	Medium	11.55668, 77.89955	30 KM/Hr
13	Turn	High	11.58214, 77.90847	15 KM/Hr
14	Turn	Medium	11.58305, 77.90733	30 KM/Hr
15	Turn	High	11.58320, 77.90646	15 KM/Hr
16	Turn	Medium	11.59323, 77.90796	30 KM/Hr
17	Turn	Medium	11.63857, 77.94831	30 KM/Hr
18	Turn	High	11.64207, 77.95199	15 KM/Hr
19	Turn	Medium	11.64708, 77.95119	30 KM/Hr
20	Turn	Medium	11.67896, 77.96882	30 KM/Hr
21	Turn	High	11.70233, 77.97869	15 KM/Hr
22	Turn	Medium	11.71394, 77.99572	30 KM/Hr
23	Turn	Medium	11.71916, 78.01117	30 KM/Hr
24	Turn	Medium	11.73786, 78.03505	30 KM/Hr
25	Turn	High	11.73652, 78.03866	15 KM/Hr
26	Turn	Medium	11.74257, 78.04708	30 KM/Hr
27	Turn	Medium	11.78039, 78.05792	30 KM/Hr
28	Turn	Medium	11.86753, 78.08331	30 KM/Hr
29	Turn	Medium	11.86981, 78.08387	30 KM/Hr
30	Turn	Medium	11.88041, 78.08143	30 KM/Hr
31	Turn	Medium	11.91302, 78.05856	30 KM/Hr
32	Turn	Medium	11.95625, 78.04748	30 KM/Hr
33	Turn	High	11.95824, 78.04981	15 KM/Hr
34	Turn	High	11.96040, 78.04947	15 KM/Hr
35	Turn	Medium	11.96320, 78.05139	30 KM/Hr
36	Turn	Medium	11.96401, 78.05134	30 KM/Hr
37	Turn	Medium	11.96463, 78.05077	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
38	Turn	Medium	11.96504, 78.04975	30 KM/Hr
39	Turn	High	11.96597, 78.04920	15 KM/Hr
40	Turn	Medium	11.96679, 78.04942	30 KM/Hr
41	Turn	Medium	11.96804, 78.05080	30 KM/Hr
42	Turn	Medium	11.98504, 78.06287	30 KM/Hr
43	Turn	Medium	12.12067, 78.13281	30 KM/Hr
44	Blind Spot	Blind Spot	12.20963, 78.11667	10 KM/Hr
45	Blind Spot	Blind Spot	12.20360, 78.12309	10 KM/Hr
46	Turn	Medium	12.21336, 78.11961	30 KM/Hr
47	Turn	High	12.23524, 78.10483	15 KM/Hr
48	Turn	High	12.23525, 78.10435	15 KM/Hr
49	Blind Spot	Blind Spot	12.37103, 78.05606	10 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
1	hospital	Omalur Government Hospital	11.74161, 78.044296	30 km/h	Medium
2	clinic	R.R. Nursing Home	11.7394129, 78.0439861	30 km/h	Medium
3	hospital	Sri Hari Hospital	11.7396726, 78.0440701	30 km/h	Medium
4	hospital	Sugumar Hospital	11.7396031, 78.0426955	30 km/h	Medium
5	hospital	Sri Sugam Hospital	11.7429821, 78.0461039	30 km/h	Medium
6	hospital	Saravana Hospital	11.7409303, 78.0458308	30 km/h	Medium
7	hospital	Arun Hospital	11.7429892, 78.0466881	30 km/h	Medium
8	clinic	Saratha Clinic	11.7422818, 78.0458175	30 km/h	Medium
10	clinic	Sri Meenakshi Nursing Home, Salem	11.7449162, 78.0474637	30 km/h	Medium
11	clinic	Divyam Child Care Clinic	11.7447135, 78.0473858	30 km/h	Medium
12	clinic	Momota Clinic	11.744982, 78.047397	30 km/h	Medium
13	clinic	Gomathi Clinic	11.7445293, 78.0470401	30 km/h	Medium
14	hospital	Vinayaka Child Hospital	11.745812, 78.0480225	30 km/h	Medium
15	hospital	Dr. G.R.Thilakam Hospital	11.746097, 78.04769	30 km/h	Medium
16	hospital	Thulasi Hospital	11.7461516, 78.0469984	30 km/h	Medium
17	hospital	Amirthalingam Hospital	11.7456708, 78.0470548	30 km/h	Medium
18	hospital	Children Hospital	11.7458377, 78.0480205	30 km/h	Medium
19	clinic	Shri Ramana Poly Clinic	11.7477122, 78.0479293	30 km/h	Medium
20	hospital	Dr. Nagaraj Hospital	11.7464035, 78.0472401	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
21	clinic	Manimuthu Skin Clinic	11.7521189, 78.0484058	30 km/h	Medium
23	hospital	Anugraha Multi Speciality Hospital	11.773549, 78.0546408	30 km/h	Medium
25	clinic	JSA Clinic	11.8700619, 78.0844982	30 km/h	Medium
26	hospital	L.R. Hospital	11.8703354, 78.0844554	30 km/h	Medium
27	hospital	Thoppur,Government Hospital	11.9398824, 78.0551658	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
9	school	Fathima Hr Sec School	11.7410508, 78.0479272	30 km/h	Medium
22	school	John Britto Matriculation Higher Secondary School	11.7676907, 78.0513092	30 km/h	Medium
24	college	Dhirajlal Gandhi College of Technology	11.7968014, 78.0562095	30 km/h	Medium
28	college	Laxminarayana Polytechnic College	12.0421277, 78.103203	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44029, 77.87544



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44900, 77.87406



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45353, 77.85697



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45781, 77.85203



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.46307, 77.84941



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45542, 77.81725



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45605, 77.81683



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.49236, 77.89552



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49365, 77.88596



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49465, 77.88488



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.55608, 77.89762



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.58320, 77.90646



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.59323, 77.90796



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.63857, 77.94831



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.64207, 77.95199



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64708, 77.95119



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.67896, 77.96882



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.70233, 77.97869



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.71394, 77.99572



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.71916, 78.01117



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.73786, 78.03505



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.73652, 78.03866



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.74257, 78.04708



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.78039, 78.05792



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.86753, 78.08331



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.86981, 78.08387



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.88041, 78.08143



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.91302, 78.05856



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.95625, 78.04748



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.95824, 78.04981



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96040, 78.04947



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96320, 78.05139



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96401, 78.05134



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96463, 78.05077



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96504, 78.04975



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96597, 78.04920



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96679, 78.04942



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96804, 78.05080



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.98504, 78.06287



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.12067, 78.13281



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 12.20360, 78.12309



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.21336, 78.11961



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.23524, 78.10483



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.23525, 78.10435



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 12.37103, 78.05606

